The fast linear operator of the Euler equations in potential-temperature form: energy-stable hybridized DGSEM with LMARS flux and relaxed trace conditions

Documentation of `hdg_relax_theta.py` / `HDGRelaxTheta.jl`

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Abstract

We derive and discretize the linear fast operator of the one-dimensional (vertical) compressible Euler equations in the variables $(\rho, \rho w, \rho\theta)$, intended as the fast (acoustic) part of a split-explicit method. Momentum advection is removed, θ is frozen in the $\rho\theta$ equation, and the pressure is written as $p = \frac{p}{\rho\theta}\,\rho\theta$ with $\frac{p}{\rho\theta}$ frozen. The resulting system is linear in the full state and has a local, weighted quadratic energy. It is discretized by a nodal discontinuous Galerkin spectral element method (DGSEM) on Gauss–Lobatto nodes in hybridized form. A split form of the volume terms, built on the summation-by-parts (SBP) property, reproduces the weighted energy identity exactly. The trace stabilization $\boldsymbol{T} = |\boldsymbol{A}|$ yields the LMARS flux, which is shown to be energy dissipative. The resulting semi-discretization is provably energy stable, conservative in ρ and $\rho\theta$, and well balanced. The plain collocation form, by contrast, has a spurious $\mathcal{O}(h)$ growth rate. The trace conditions are relaxed into evolution equations, and the coupled system is integrated without static condensation.

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1 Model and linearization

1.1 Full equations

In a vertical column with $m = \rho w$ and $\Theta = \rho\theta$,

$$\partial_t \rho + \partial_z m = 0, \qquad \partial_t m + \partial_z(mw + p) = -\rho g, \qquad \partial_t \Theta + \partial_z(\theta m) = 0, \qquad p = p_{00}\Big(\frac{R\,\Theta}{p_{00}}\Big)^{\gamma}. \tag{1}$$

1.2 Linearization (fast operator)

Given a frozen state (bars), for example the current slow state, the linearization consists of three steps.

1. Remove momentum advection: $\partial_z(mw) \to 0$.
2. Freeze θ in the Θ equation: $\partial_z(\theta m) \to \partial_z(\bar\theta\, m)$.
3. Write $p = \dfrac{p}{\Theta}\,\Theta$ and freeze the ratio: $p \to \bar\pi\,\Theta$ with $\bar\pi = \bar p/\bar\Theta = R\bar\Pi$ (Π = Exner function).

This gives

$$\boxed{\partial_t \rho + \partial_z m = 0, \qquad \partial_t m + \partial_z(a\,\Theta + b) = -g\rho, \qquad \partial_t \Theta + \partial_z(\bar\theta\, m) = 0,} \tag{2}$$

with $a = \texttt{pfac}\,\bar\pi$ and $b = (1 - \texttt{pfac})\,\bar p$. The default $\texttt{pfac} = 1$ is the linearization above. The system is linear in the *full* state $\boldsymbol{q} = (\rho, m, \Theta)$. In flux form, $\partial_t \boldsymbol{q} + \partial_z(\boldsymbol{A}\boldsymbol{q} + b\,\boldsymbol{e}_m) = \boldsymbol{S}\boldsymbol{q}$ with

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & a \\ 0 & \bar\theta & 0 \end{pmatrix}, \qquad \boldsymbol{S} = \begin{pmatrix} 0 & 0 & 0 \\ -g & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \operatorname{spec}\boldsymbol{A} = \{0, \pm c\}, \quad c^2 = a\bar\theta. \tag{3}$$

Sound speed. For $\texttt{pfac} = 1$, $c^2 = \bar\pi\bar\theta = \bar p/\bar\rho = R\bar T$: freezing p/Θ gives the isothermal sound speed (287 instead of 340 m/s at 288 K). With $\texttt{pfac} = \gamma$ one obtains the tangent-linear pressure and $c^2 = \gamma R\bar T$; this variant coincides with the classical linearization about a hydrostatic state at rest. The constant b keeps $p_{\mathrm{lin}}(\bar\Theta) = \bar p$.

1.3 Use in a split-explicit method

With the full operator $\mathcal{N}$ and the fast operator $\mathcal{L}$ frozen at the slow level $\boldsymbol{q}^n$, the fast system over the small steps is $\partial_t \boldsymbol{q} = \mathcal{L}(\boldsymbol{q}) + \boldsymbol{F}_{\mathrm{slow}}$ with $\boldsymbol{F}_{\mathrm{slow}} = \mathcal{N}(\boldsymbol{q}^n) - \mathcal{L}(\boldsymbol{q}^n)$ held constant (e.g. [6]). All time integrators accept `F_slow`, and `tendency(q)` returns $\mathcal{L}(\boldsymbol{q})$.

2 Energy of the frozen system

Let $p = a\Theta$ and let $\chi = \Theta - \bar\theta\rho$ be the θ-perturbation. From (2),

$$\partial_t \chi = -\partial_z(\bar\theta m) + \bar\theta\,\partial_z m = -\bar\theta_z\, m. \tag{4}$$

Proposition 1 (Energy). *Let $\varphi(z) = \bar\theta(z)\exp\big(\int^z g/c^2\,dz'\big)$. For deviations from a steady state,*

$$E = \frac{1}{2}\int \varphi\Big[m^2 + \frac{p^2}{c^2} + \frac{g}{\bar\theta\,\bar\theta_z}\,\chi^2\Big]dz, \qquad \frac{dE}{dt} = -\big[\varphi\, p\, m\big]_{\partial\Omega}. \tag{5}$$

E is positive definite if the frozen stratification is stable ($\bar\theta_z > 0$).

Proof. Write $\rho = (\Theta - \chi)/\bar{\theta}$ in the momentum equation. Testing the momentum equation with φm and (4) with $\varphi g \chi/(\bar{\theta}\bar{\theta}_z)$ gives two terms $\pm\varphi g m \chi/\bar{\theta}$, which cancel. Testing the Θ equation with $\varphi p/c^2 = \varphi\Theta/\bar{\theta}$ and integrating by parts gives $-[\varphi p m] + \int m\,\partial_z(\varphi p) - \int \varphi p\,\bar{\theta}_z m/\bar{\theta}$. Together with the momentum term $-\int \varphi m(\partial_z p + g\Theta/\bar{\theta})$, the remainder is $\int m\,p\,(\varphi_z - \varphi\bar{\theta}_z/\bar{\theta} - \varphi g/c^2)$, which vanishes by the choice of φ. □

The energy is local in (ρ, m, Θ) but *weighted*, and it couples ρ and Θ through χ. A Galerkin method with unweighted test functions does not reproduce (5). This is the origin of the instability of the standard form (Section 5).

3 Hybridized DGSEM discretization

3.1 Nodal basis and SBP property

Each element K_e (size h, Jacobian $J = h/2$) carries the Lagrange polynomials on the $p+1$ Gauss–Lobatto–Legendre (GLL) nodes ξ_i with weights w_i. The unknowns are nodal values, and quadrature is collocated at the nodes. The mass matrix is $\boldsymbol{M} = J\,\boldsymbol{W} = J\,\mathrm{diag}(w_i)$, and the differentiation matrix $D_{ij} = \ell_j'(\xi_i)$ satisfies the SBP property [3, 2]

$$\boldsymbol{W}\boldsymbol{D} + \boldsymbol{D}^{\mathsf{T}}\boldsymbol{W} = \boldsymbol{B} = \mathrm{diag}(-1, 0, \ldots, 0, 1). \tag{6}$$

Products of the frozen coefficients with the unknowns are evaluated pointwise at the nodes. The element end nodes coincide with the faces, so the face values are just the end-node values.

3.2 Split-form volume terms

At node i (all quantities nodal, $(\boldsymbol{D}u)_i = \sum_j D_{ij}u_j$):

$$\dot{\rho}_i = -\tfrac{1}{J}(\boldsymbol{D}m)_i \;+\; \mathrm{SAT}, \tag{7}$$

$$\dot{\Theta}_i = -\tfrac{1}{J}\big(\bar{\theta}_i(\boldsymbol{D}m)_i + m_i(\boldsymbol{D}\bar{\theta})_i\big) \;+\; \mathrm{SAT}, \tag{8}$$

$$\dot{m}_i = -\tfrac{1}{J}\Big[\tfrac{1}{\varphi_i}\big(\boldsymbol{D}(\varphi a\Theta)\big)_i - \tfrac{a_i\Theta_i}{\bar{\theta}_i}(\boldsymbol{D}\bar{\theta})_i + (\boldsymbol{D}b)_i\Big] + \frac{g\,\Theta_i}{\bar{\theta}_i} - g\rho_i \;+\; \mathrm{SAT}. \tag{9}$$

- The Θ flux is in product-rule form. By SBP, $\sum_i w_i[\bar{\theta}_i(\boldsymbol{D}m)_i + m_i(\boldsymbol{D}\bar{\theta})_i] = [\bar{\theta}m]_{-1}^{1}$, so Θ is still exactly conserved. Moreover, $\dot{\chi}_i = -(\boldsymbol{D}\bar{\theta})_i m_i/J$ holds exactly at every node, which is the discrete version of (4).
- The pressure gradient (9) is consistent with $\partial_z(a\Theta + b)$, because $\varphi^{-1}\partial_z(\varphi p) - p\bar{\theta}_z/\bar{\theta} - g\Theta/\bar{\theta} = \partial_z p$ by the definition of φ. It is the form for which the discrete energy identity holds.
- The nodal values φ_i are computed from $\bar{\theta}$ and an accurate quadrature of $\int g/c^2$. They are continuous across faces.

3.3 Traces, SAT and the LMARS flux

Let b be an end node of an element with outward normal n, and f its face. The coupling to the trace is

$$J\,w_b\,\dot{\boldsymbol{q}}_b \mathrel{+}= (\boldsymbol{A}_f n - \boldsymbol{T}_f)(\boldsymbol{q}_b - \hat{\boldsymbol{q}}_f), \qquad \boldsymbol{T}_f = |\boldsymbol{A}_f| = \begin{pmatrix} 0 & 0 & a_f/c_f \\ 0 & c_f & 0 \\ 0 & 0 & c_f \end{pmatrix}. \tag{10}$$

This is the strong-form SAT with the hybridized flux $\widehat{\boldsymbol{F}}\cdot n = \boldsymbol{A}_f\hat{\boldsymbol{q}}\,n + \boldsymbol{T}_f(\boldsymbol{q}_b - \hat{\boldsymbol{q}})$. The trace conditions are $\hat{\boldsymbol{q}}_f = \{\{\boldsymbol{q}\}\}$ at interior faces and $\hat{\boldsymbol{q}}_f = \mathrm{diag}(1,0,1)\,\boldsymbol{q}_{\mathrm{int}}$ at walls ($\hat{m} = 0$). With them, the face flux is exactly LMARS [1]:

$$m^* = \{\{m\}\} - \frac{[\![p]\!]}{2c_f}, \qquad p^* = \{\{p\}\} + b_f - \frac{c_f}{2}[\![m]\!], \qquad \Theta\text{ flux} = \bar{\theta}_f\,m^*, \tag{11}$$

with $p = a\Theta$. Since $c_f^2 = a_f \bar{\theta}_f$, the Θ flux equals $\bar{\theta}_f$ times the mass flux.

Proposition 2 (Discrete energy stability)**.** *Let $\Delta \boldsymbol{q}$ be the deviation from a discrete steady state, and let $\bar{\theta}$ be stably stratified at the nodes ($(\boldsymbol{D}\bar{\theta})_i > 0$). For the split form with LMARS traces and $\varepsilon = 0$, the discrete energy*

$$E_h = \frac{1}{2} \sum_e J \sum_i w_i \varphi_i \Big[m_i^2 + \frac{a_i}{\bar{\theta}_i} \Theta_i^2 + \frac{g}{\bar{\theta}_i (\boldsymbol{D}\bar{\theta})_i / J} \chi_i^2 \Big]$$

satisfies

$$\frac{dE_h}{dt} = - \sum_{\text{interior } f} \varphi_f \Big(\frac{c_f}{2} [\![m]\!]^2 + \frac{1}{2c_f} [\![p]\!]^2 \Big) - \sum_{\text{walls}} \varphi_f \, c_f \, m_b^2 \;\le\; 0.$$

Sketch. The proof mirrors Proposition 1, with SBP replacing integration by parts:

- The χ terms cancel pointwise, because $\dot{\chi}_i = -(\boldsymbol{D}\bar{\theta})_i m_i / J$ holds exactly.
- The volume terms reduce to $-[\varphi p m]$ at each element's ends. This uses $\sum_i w_i \psi_i (\boldsymbol{D}m)_i = [\psi m] - \sum_i w_i m_i (\boldsymbol{D}\psi)_i$ with $\psi = \varphi a \Theta / \bar{\theta}$.

At each element end, the volume boundary term and the SAT contributions combine to $-\varphi\,[m\,p^* + p\,m^* - p\,m]\,n$. The χ parts of the SAT cancel because the Θ flux is $\bar{\theta}_f$ times the mass flux. Summing the two sides of an interior face and inserting (11) gives $\varphi_f(p^*[\![m]\!] + m^*[\![p]\!] - [\![pm]\!]) = -\varphi_f(\frac{c_f}{2}[\![m]\!]^2 + \frac{1}{2c_f}[\![p]\!]^2)$. At walls, $m^* = 0$ and $p^* = p_b + c_f m_b n$. □

Remark 1. *Without the split form (plain collocation of the weak form with fluxes $a\Theta$ and $\bar{\theta}m$), the identity is violated by $\mathcal{O}(h)$ terms. The violation is independent of p and of the penalty, and it produces a real positive eigenvalue proportional to h (Section 5). The same holds for a modal discretization with exact quadrature: the problem is the unweighted Galerkin test, not aliasing.*

3.4 Well-balancing

A discrete hydrostatic state ($m = 0$, continuous nodal Θ) has vanishing ρ- and Θ-tendencies, and the momentum residual is cancelled by $\rho_i = -\big(\text{pressure part of (9)}\big)_i / g$. The code computes this ρ from the assembled operator (`balanced_state`). It is steady to round-off for every ε.

For `pfac` $= 1$ there is more. The analytic background from which the coefficients are frozen satisfies $\varphi\bar{p} = p_{00}\,\bar{\theta}$, because $\bar{p}_z/\bar{p} = -g/c^2$. Its nodal interpolant is therefore *exactly* balanced: $\varphi^{-1}\boldsymbol{D}(\varphi\bar{p}) = \bar{p}\,\boldsymbol{D}\bar{\theta}/\bar{\theta}$, and (9) reduces to $g(\bar{\Theta}/\bar{\theta} - \bar{\rho}) = 0$. The weight φ is the integrating factor of the discrete hydrostatic equation.

3.5 Matrix form and relaxation

With all nodal values in $\boldsymbol{q}$ and traces in $\hat{\boldsymbol{q}}$,

$$\begin{pmatrix} \boldsymbol{M} & \boldsymbol{0} \\ \boldsymbol{0} & \varepsilon \boldsymbol{I} \end{pmatrix} \frac{d}{dt} \begin{pmatrix} \boldsymbol{q} \\ \hat{\boldsymbol{q}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{K} & \boldsymbol{G} \\ \boldsymbol{H} & -\boldsymbol{I} \end{pmatrix} \begin{pmatrix} \boldsymbol{q} \\ \hat{\boldsymbol{q}} \end{pmatrix} + \begin{pmatrix} \boldsymbol{f}_b + \boldsymbol{M}\boldsymbol{F}_{\text{slow}} \\ \boldsymbol{0} \end{pmatrix}, \qquad \boldsymbol{M} = \operatorname{diag}(J w_i). \tag{12}$$

For $\varepsilon = 0$ this is the HDG/LMARS scheme. For $\varepsilon > 0$ the trace condition $\hat{\boldsymbol{q}} = \boldsymbol{H}\boldsymbol{q}$ is relaxed. The element unknowns are not eliminated. The relaxation error is $\mathcal{O}(\varepsilon c/h)$. The mass matrix is diagonal, so the explicit stages need no solves at all.

4 Time integration

- `method="dirk"`: five-stage, fourth-order, L-stable, stiffly accurate SDIRK of Hairer–Wanner [5]. It uses one sparse LU of the full coupled matrix and is valid for every $\varepsilon \geq 0$, so it can serve as a vertically implicit (HEVI-type) option.
- `method="imex"`: ARS(2,2,2) [4], with the element part explicit and the relaxation implicit. The implicit stage is local per face, $\hat{\boldsymbol{q}}^{(i)} = (\hat{\boldsymbol{q}}^{*} + k\boldsymbol{H}\boldsymbol{q}^{(i)})/(1+k)$ with $k = \Delta t\, a^{I}_{ii}/\varepsilon$. As $\varepsilon \to 0$ it becomes an explicit DGSEM/LMARS step under the CFL condition $\Delta t \lesssim h/(c(2p+1))$. This is the natural small-step integrator.

5 Verification

All numbers come from `run_tests_theta.py`. The Julia port prints the same tests, with these values in its comments.

5.1 Stability

Table 1: Largest real part of the spectrum of the semi-discrete operator ($\varepsilon = 0$, walls, $[0, 10]$ km, $p = 3$).

atmosphere, `pfac`	form	$N_e = 8$	16	32
isothermal, 1	split	2.4e-16	8.8e-16	3.2e-15
	standard	7.2e-4	3.6e-4	1.8e-4
isothermal, γ	split	4.3e-16	1.2e-15	2.3e-15
	standard	6.1e-4	3.1e-4	1.5e-4
lapse 6.5 K/km, 1	split	3.4e-16	8.6e-16	2.8e-15
	standard	3.5e-4	1.8e-4	8.8e-5
lapse 6.5 K/km, γ	split	7.0e-16	7.6e-16	2.6e-15
	standard	2.9e-4	1.5e-4	7.4e-5

For the split form, the symmetric part of $\mathcal{P}\boldsymbol{L}$ ($\mathcal{P}$ the energy matrix of Proposition 2) is negative semidefinite to round-off. The relaxed system ($\varepsilon c/h = 10^{-3}, 10^{-1}, 1$) also has no eigenvalue with positive real part ($\leq 8 \cdot 10^{-14}$).

A random perturbation of the isothermal balanced state ($N_e = 16$, $p = 3$, DIRK, $\Delta t = 20$ s) gives, after 6 h, $E(6\,\mathrm{h})/E(0) = 0.79$ for the split form and $1.2 \cdot 10^{6}$ for the standard form. The standard form's growth rate is real and proportional to h, independent of p and of the penalty strength. Its growing mode is grid scale: m alternates in sign from element to element, so $\hat{m} \approx 0$ and the elements act as weakly closed boxes.

5.2 Convergence

Constant frozen coefficients ($\bar{\pi} = R$, $\bar{\theta} = 300$ K), periodic domain of 2000 m, exact normal mode, one acoustic transit. Initial data are nodal interpolants. DGSEM is less accurate than the modal Galerkin version at equal p, because the GLL quadrature is inexact (factor ≈ 2 at $p = 3$), but it has the same order. The standard form has the same accuracy as the split form.

5.3 Well-balancing and slow forcing

- **Discrete hydrostatic state** (lapse-rate atmosphere, walls, $N_e = 40$, $p = 3$): after 200 steps the relative deviation is $\approx 10^{-12}$ with DIRK and $\approx 10^{-15}$ with IMEX, for `pfac` $= 1$ and γ and $\varepsilon = 0$ and 10^{-3}.

Table 2: L^2 error in Θ (rates for ρ, m, Θ coincide).

method	p	$N_e = 8$	16	32	64	rate
DIRK, $\varepsilon = 0$	1	5.58e+0	1.82e+0	4.92e-1	1.26e-1	1.97
	2	9.21e-2	9.84e-3	1.17e-3	1.44e-4	3.02
	3	3.13e-3	1.96e-4	1.22e-5	7.66e-7	4.00
DIRK, $\varepsilon = 10^{-9}h/c$	3	3.13e-3	1.96e-4	1.23e-5	7.82e-7	3.97
IMEX, $\varepsilon = 10^{-9}h/c$	1	5.56e+0	1.81e+0	4.90e-1	1.25e-1	1.97
	2	8.52e-2	9.13e-3	1.27e-3	2.33e-4	2.45

Table 3: L^2 error in Θ versus $\varepsilon c/h$ ($N_e = 16$, $p = 3$; $\varepsilon = 0$: $1.96 \cdot 10^{-4}$ DIRK, $1.64 \cdot 10^{-3}$ IMEX).

$\varepsilon c/h$	10^{-8}	10^{-7}	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}
DIRK	1.96e-4	1.97e-4	2.52e-4	1.57e-3	1.55e-2	1.55e-1	1.52e+0
IMEX	1.64e-3	1.62e-3	1.48e-3	2.15e-4	1.39e-2	1.53e-1	1.52e+0

- **Interpolated analytic background:** with `pfac` $= 1$, $\max|\dot{\boldsymbol{q}}| \approx 10^{-12}$ for $N_e = 10, \ldots, 80$, i.e. exact balance. With `pfac` $= \gamma$, $\max|\dot{\boldsymbol{q}}| = 1.6 \cdot 10^{-5}, \ldots, 3.2 \cdot 10^{-8}$, converging at rate 3.
- **Slow forcing:** with $\boldsymbol{F}_{\text{slow}} = -\mathcal{L}(\boldsymbol{q}^*)$ the state stays at $\boldsymbol{q}^*$ to $4 \cdot 10^{-11}$ (DIRK) and 10^{-17} (IMEX).

5.4 Acoustic pulse between walls

Discrete hydrostatic background plus a 100 Pa pressure pulse at 3 km, $N_e = 80$, $p = 3$, $t = 25$ s. Mass drift: $2.3 \cdot 10^{-11}$ (`pfac` $= 1$, $\varepsilon = 0$), $3.3 \cdot 10^{-10}$ (`pfac` $= \gamma$), and $1.4 \cdot 10^{-5}$ for $\varepsilon = 10^{-4}h/c$; the totals are $\sim 10^4\,\mathrm{kg\,m^{-2}}$. The weighted energy never increased.

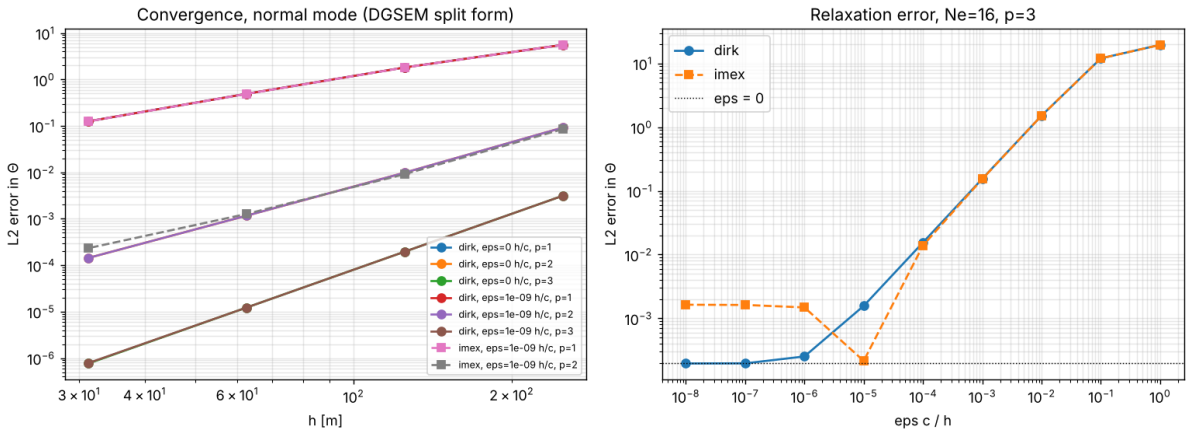


Figure 1: Left: convergence. Right: relaxation error versus $\varepsilon c/h$.

6 Code

`hdg_relax_theta.py`	Python solver (NumPy/SciPy)
`run_tests_theta.py`	Python tests and plots
`julia/HDGRelaxTheta.jl`	Julia module (standard library only)
`julia/run_tests_theta.jl`	Julia tests, writes CSV files

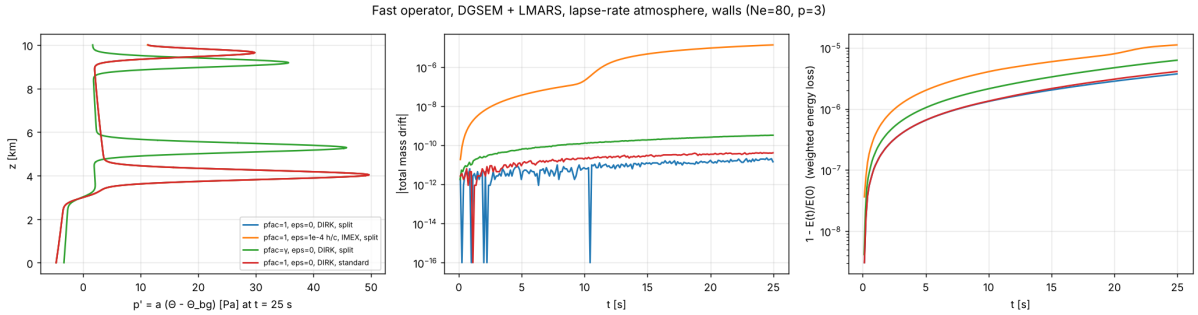


Figure 2: Pulse at $t = 25\,\mathrm{s}$ (left), mass drift (middle), and loss of weighted energy $1 - E/E_0$ (right).

Data layout

Nodal value (e, r, i) (element, variable ρ, m, Θ, GLL node from bottom to top) is stored at $((e-1)\cdot 3 + (r-1))(p+1) + i$ in Julia (1-based) and at $(3e + r)(p+1) + i$ in Python (0-based). Trace (f, r) is stored at $3(f-1) + r$ (Julia) or $3f + r$ (Python). The state is $\boldsymbol{U} = (\boldsymbol{q}, \hat{\boldsymbol{q}})$.

Usage (Julia)

```
include("HDGRelaxTheta.jl"); using .HDGRelaxTheta
atm = Atmosphere(T0 = 288.0, Γ= 0.0065)
fr = frozen_from_atmosphere(atm; pfac = 1.0) # or FrozenState(pi_fun, theta_fun; ...)
s = HDGRelaxTh(80, 3, 0.0, 1.0e4, fr; bc = :wall, eps = 0.0, form = :split)
q = balanced_state(s, z -> atm_Theta(atm, z)) # discrete hydrostatic state
F = zeros(s.nq) # slow tendency (split-explicit)
U = run!(s, initial_state(s, q), 0.05, 100; method = :imex, F_slow = F)
z, Q = evaluate(s, U[1:s.nq]) # Q[1,:]=ρ, Q[2,:]=ρw, Q[3,:]=ρθ
```

Python:

```
s = HDGRelaxTheta(Ne, p, zmin, zmax, FrozenState.from_atmosphere(atm, pfac),
                  bc="wall", eps=0.0, form="split")
U = s.run(U0, dt, n, method="imex", F_slow=F)
```

To freeze coefficients from a DG slow state, pass functions that return its continuous nodal values of $\bar{p}/\bar{\Theta}$ and $\bar{\theta}$. The energy weight φ is built internally.

References

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